

Correction Model — Technical Documentation

Generated from `bear/correction_features.csv` and `bear/targets.csv`. Reproduce with `python -m bear.make_docs`.

1. Overview

- **Objective:** estimate a binary **6-month rolling correction**, $y_t = 1\{\text{MDD}_t \leq -10\%\}$, i.e. any drawdown deeper than 10% over the rolling 6-month forward window.
- **Model:** weight-constrained binary logistic on standardized factors, signs fixed to economic priors, calibrated to the base rate.
- **Horizon:** 6 months (rolling forward window).
- **Sample:** 1960-01-31 → 2026-06-30 (797 complete monthly observations).
- **Historical base rate:** 34.9%.
- **Current reading (2026-06-30):** P(event) = **29.4%**.
- **Inference:** Newey-West HAC, max lag = 6 months.

2. Feature Engineering

All raw series are monthly (daily/weekly series are sampled to month-end), shifted forward by their real-world publication lag to prevent look-ahead, and transformed to stationary form. Each factor is then standardized on the training sample, $\tilde{x}_{i,t} = (x_{i,t} - \mu_i) / \sigma_i$.

#	Factor	Raw series	Transformation	Rationale
1	<code>vtc_slope</code>	VIX term structure (VIX3M – VIX)	Level (slope)	Steep contango = complacency that often precedes corrections; backwardation = stress.
2	<code>spx_vs_10ma</code>	S&P 500 price (monthly close)	% deviation from 10-month MA ($\approx$ 200-day)	Trend deterioration; price below the long MA is risk-off (Moskowitz-Ooi-Pedersen 2012).
3	<code>m12_1_mom</code>	S&P 500 price (monthly close)	12-1 momentum (12m return excluding last month)	Stretched momentum raises mean-reversion correction risk.
4	<code>anfci_3m_chg</code>	Chicago Fed Adjusted NFCI (FRED ANFCI)	3-month change	Tightening financial conditions (orthogonalized to the economy) precede pullbacks.
5	<code>cape_20yr_pct</code>	Shiller CAPE (P/E10)	20-year trailing percentile	Valuation extreme — a severity conditioner; weak standalone timer (Goyal-Welch 2008).
6	<code>baa_zscore_24m</code>	Moody's BAA – 10y default spread (FRED BAA10Y)	24-month trailing z-score	Fast credit-stress signal (the Tokic-Jackson correction→bear bridge).

Forward drawdown target (rolling window):

$$\text{MDD}_t = \min_{t < u \leq t+6} \left(\frac{P_u}{\max_{t < v \leq u} P_v} - 1 \right)$$

3. Constrained Logistic Model

Link / specification:

$$\hat{p}_t = \Pr(y_t = 1) = \sigma(z_t) = \frac{1}{1 + e^{-z_t}}$$

$$z_t = \beta_0 + \sum_{i=1}^6 \beta_i \tilde{x}_{i,t}$$

Estimation: coefficients are written as $\beta_i = \text{sign}_i \cdot \gamma_i$ with $\gamma_i \geq 0$ (signs fixed to economic priors), fitted by maximizing the Bernoulli likelihood subject to per-factor weight bounds:

$$w_i = \frac{|\beta_i|}{\sum_j |\beta_j|}, \quad 0\% \leq w_i \leq 30\%$$

Fitted model:

$$z_t = -0.653 + 0.210 \tilde{x}_1 - 0.418 \tilde{x}_2 + 0.286 \tilde{x}_3 + 0.074 \tilde{x}_4 - 0.144 \tilde{x}_5 + 0.373 \tilde{x}_6$$

Fitted coefficients (sorted by weight):

Factor	Coefficient β_i	Weight w_i	p (HAC)
spx_vs_10ma	-0.4182	27.8%	0.051 .
baa_zscore_24m	+0.3727	24.8%	0.057 .
m12_1_mom	+0.2859	19.0%	0.138
vts_slope	+0.2095	13.9%	0.360
cape_20yr_pct	-0.1443	9.6%	0.360
anfci_3m_chg	+0.0742	4.9%	0.541
intercept	-0.6529	—	—

* p<0.05 · . p<0.10 (Newey-West HAC, max lag = 6).

4. Regression Performance

4.1 Specification

- **Form:** binary logistic, identity-of-weights constraint $w_i \in [0\%, 30\%]$, 6 factors, signs fixed.

- **Autocorrelation:** the 6-month rolling target overlaps across consecutive months (consecutive observations share 5/6 of their window). Standard errors are Newey-West **HAC**-corrected with a Bartlett kernel and max lag = 6 months.
- **Calibration:** fitted by natural-weight likelihood, so the output is calibrated to the 34.9% base rate.

4.2 Sensitivity

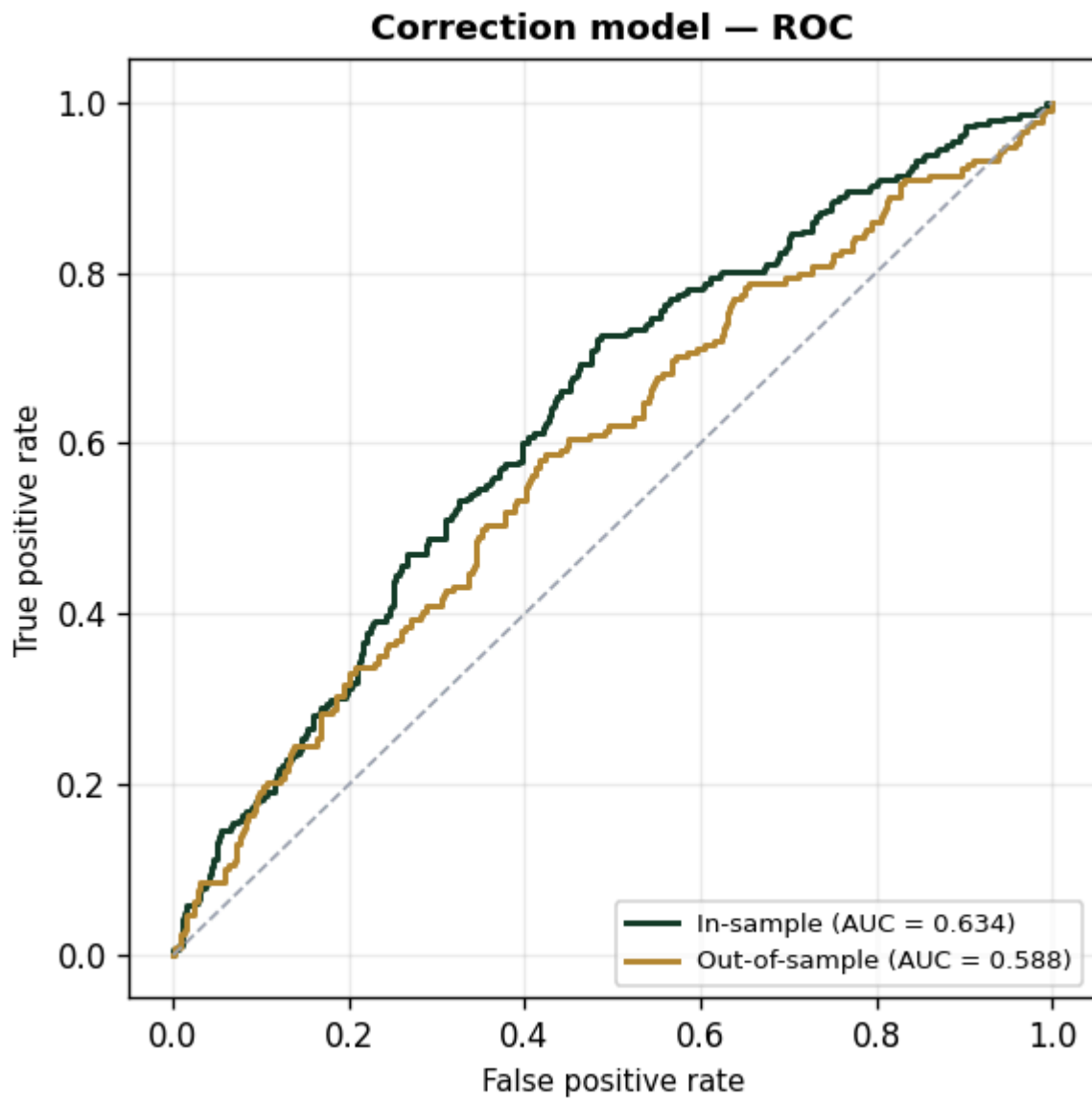
Marginal effect of a **+1 standard-deviation** move in each factor on the model output, evaluated at the current reading ($\partial \hat{y} = \hat{y}(1 - \hat{y}) \beta_i$, $\hat{y} = 0.294$):

Factor	β_i	Weight	Δ output / +1 SD	p (HAC)	Push
S&P 500 vs 10-month MA	-0.4182	27.8%	-0.0867 (-8.67 pp)	0.051	Bullish
BAA spread, 2yr z-score	+0.3727	24.8%	+0.0773 (+7.73 pp)	0.057	Bullish
12-1 month price momentum	+0.2859	19.0%	+0.0593 (+5.93 pp)	0.138	Bearish
VIX term structure slope (VIX3M-VIX)	+0.2095	13.9%	+0.0434 (+4.34 pp)	0.360	Bearish
Shiller CAPE, 20yr percentile	-0.1443	9.6%	-0.0299 (-2.99 pp)	0.360	Bullish
Adj. financial conditions, 3m change	+0.0742	4.9%	+0.0154 (+1.54 pp)	0.541	Bearish

4.3 Area Under the Curve (AUC)

Discrimination is measured against the realized event (*Correction event: 6-month drawdown > 10%*); the predicted probability is the ranking score.

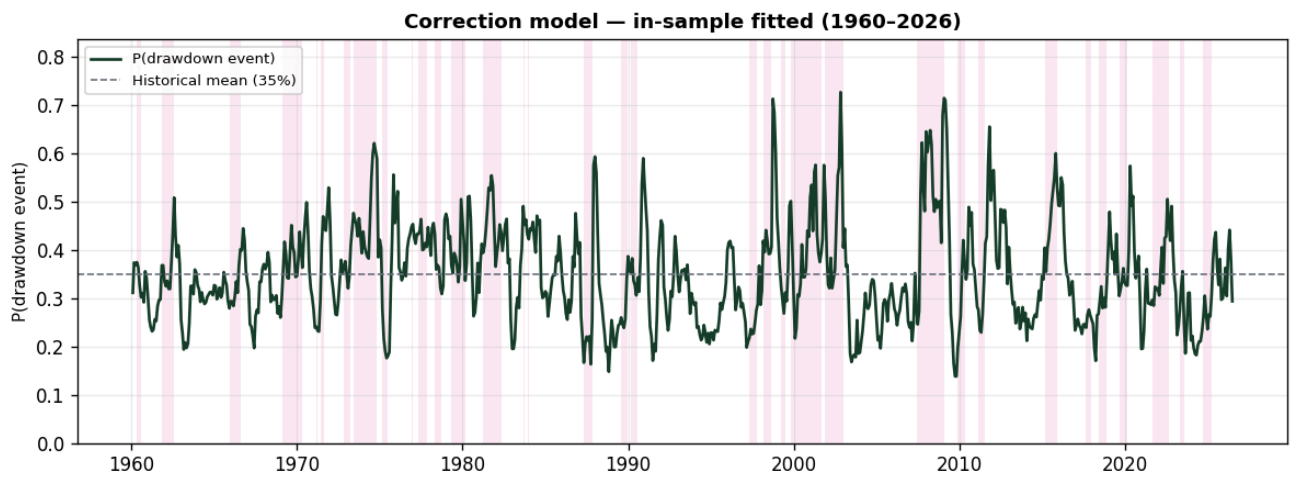
Sample	AUC	N	Events
In-sample	0.634	797	278
Out-of-sample (walk-forward)	0.588	617	208



5. Charts

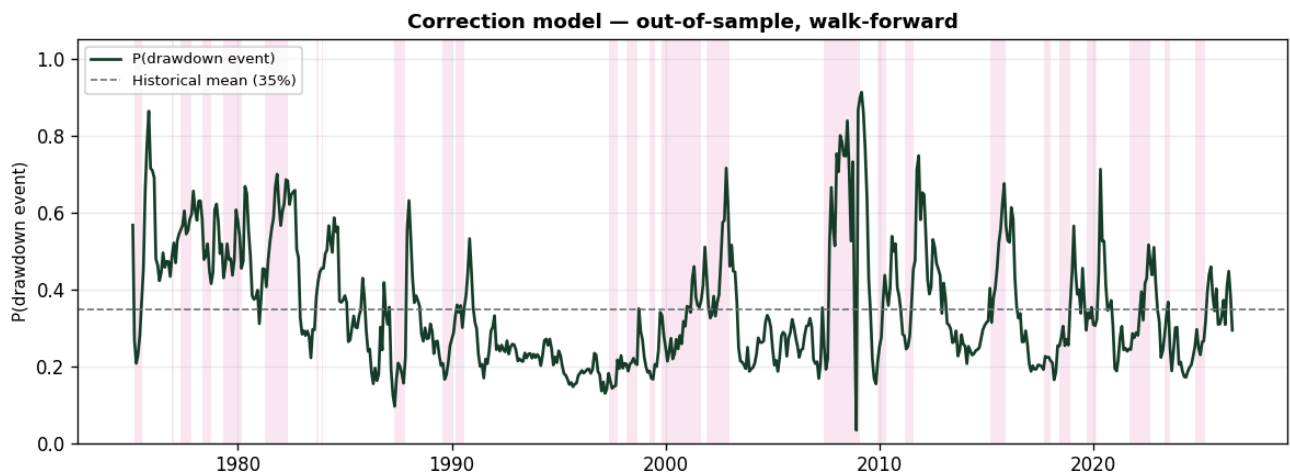
5.1 In-sample fit

Fitted values across the full sample (parameters estimated on the full sample). Shaded spans mark the realized rolling-window event.



5.2 Out-of-sample (walk-forward)

Expanding-window estimate: at each month the model is re-fit on prior data only, then predicts that month (no look-ahead).



6. Appendix — Realized Correction Events

The 278 event-months in the AUC sample (signal months whose 6-month forward drawdown exceeded 10%) group into **35 distinct episodes**. *Start/End* are the first and last signal months of each run; *Worst drawdown* is the deepest 6-month forward drawdown observed during the episode.

#	Start	End	Signal months	Worst drawdown
1	1960-04	1960-07	4	-10.1%
2	1961-10	1962-07	10	-26.4%
3	1965-11	1966-07	9	-19.5%
4	1967-09	1967-09	1	-10.0%
5	1969-01	1970-04	16	-25.9%
6	1971-02	1971-03	2	-10.7%
7	1971-05	1971-08	4	-11.0%
8	1972-10	1973-03	6	-15.0%
9	1973-05	1974-10	18	-33.1%
10	1975-02	1975-06	5	-14.1%
11	1976-11	1976-12	2	-10.4%
12	1977-04	1977-10	7	-11.2%
13	1978-04	1978-09	6	-13.6%
14	1979-04	1980-02	11	-17.1%
15	1981-03	1982-04	14	-17.4%
16	1983-08	1983-09	2	-10.6%
17	1983-11	1983-12	2	-12.0%
18	1987-04	1987-10	7	-33.5%
19	1989-07	1989-12	6	-10.2%
20	1990-02	1990-07	6	-19.9%
21	1997-04	1997-09	6	-10.8%
22	1998-02	1998-08	7	-19.3%
23	1999-03	1999-06	4	-12.1%
24	1999-10	2001-08	23	-26.4%
25	2001-11	2002-12	14	-31.8%
26	2007-05	2009-01	21	-46.4%
27	2009-11	2010-04	6	-16.0%
28	2011-02	2011-07	6	-19.2%
29	2015-02	2015-11	10	-13.3%
30	2017-08	2017-12	5	-10.2%
31	2018-05	2018-11	7	-19.8%
32	2019-08	2020-02	7	-33.9%

#	Start	End	Signal months	Worst drawdown
33	2021-08	2022-08	13	-23.6%
34	2023-04	2023-07	4	-10.3%
35	2024-09	2025-03	7	-18.9%
		Total	278	